

# Qatar-2b

R-0764

Benchmark run · workflow W8-benchmark-deep · record deep-bench\_Qatar-2\_240413 · created 2026-07-28T04:22:11+00:00

BENCHMARK: ACCURATE — fit consistent with published values

## Results

|                                                   |                               |
|---------------------------------------------------|-------------------------------|
| Mid-Transit Time (Tmid)                           | 2460413.82 +/- 0.0013 BJD_TDB |
| Ratio of Planet to Stellar Radius (Rp/R*)         | 0.177 +/- 0.014               |
| Transit depth (Rp/Rs)^2                           | 3.12 +/- 0.51 [%]             |
| Orbital Inclination (inc)                         | 89.71 +/- 1.15                |
| Airmass coefficient 1 (a1)                        | 1.188 +/- 0.019               |
| Airmass coefficient 2 (a2)                        | -0.122 +/- 0.036              |
| Scatter in the residuals of the lightcurve fit is | 1.62 %                        |
| Best Comparison Star                              | #2 - [614, 120]               |
| Optimal Aperture                                  | 3.16                          |
| Optimal Annulus                                   | 3.93                          |
| Transit Duration (day)                            | 0.0776 +/- 0.0012             |

## Accuracy vs published ephemeris (ExoClock/NEA)

|                                      |                                                         |
|--------------------------------------|---------------------------------------------------------|
| Rp/R $\blacksquare$ fit vs published | 0.177 $\pm$ 0.014 vs 0.16327 (deviation 0.98 $\sigma$ ) |
| Duration fit vs published            | 0.0776 d vs 0.0767 d (deviation 0.78 $\sigma$ )         |
| Mid-time O-C                         | 2.1 min                                                 |
| Depth SNR                            | 12.6                                                    |
| Residual scatter                     | 1.62 %                                                  |

## Light curve

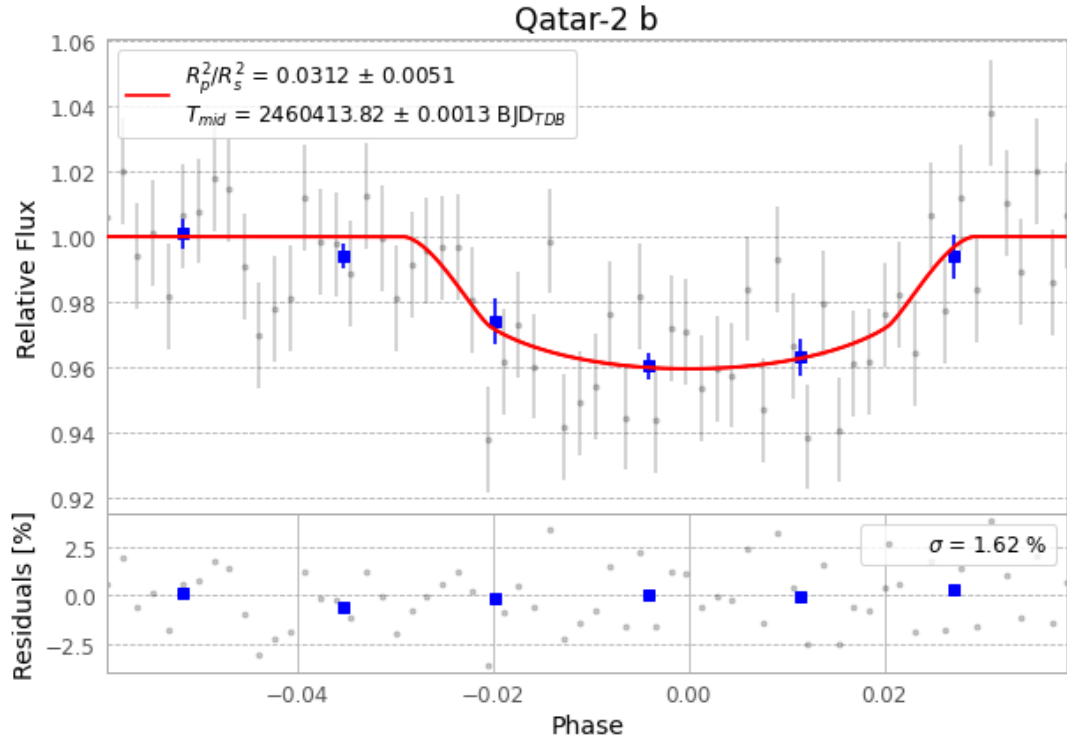


Figure 1 — FinalLightCurve\_Qatar-2 b\_2024-04-12.png (SHA-256 2f756e89f67e96b6...)

## Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [517, 131], 3 comparison star(s). Exact configuration: parameters.inits\_json in record.json (SHA-256 aa8e66a1dfcdf4c7...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline\_commit ead60b9

## Data provenance

64 FITS frames (42 MB), Qatar-2240413053616.FITS → Qatar-2240413084516.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Qatar-2-240413.log (631d048cac4d...), FinalLightCurve\_Qatar-2 b\_2024-04-12.png (2f756e89f67e...), FinalLightCurve\_Qatar-2 b\_2024-04-12.csv (8553a8f057ec...)

## Compute resources

Wall time 110.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

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Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-28 04:22Z.